

Mohamed ELRady

AI engineer

✉ mohamedelrady554@gmail.com ☎ 01112021948 | 01091418618 📍 Egypt 📅 Apr 23,2004

🐙 GitHub 🌐 LinkedIn 🗣️ WhatsApp

SUMMARY

AI Engineer with hands-on experience building end-to-end applications using computer vision, multimodal models, LLM agents, and machine learning. Skilled in Python, PyTorch, scikit-learn, pandas, YOLO, CLIP, Whisper, TypeScript, and React. Developed bilingual Arabic–English solutions for video analysis, business intelligence, prompt optimization, health education, and large-scale data mining, with an emphasis on grounded outputs, privacy, testing, and measurable performance.

EDUCATION

Beni-Suef National University

09/2023 – Present

B.Sc. in Computer Science & Artificial Intelligence (AI and Data Science Track)

Beni-Suef

- Expected Graduation: July 2027

EXPERIENCE

AI & Machine Learning Engineer

National Telecommunication Institute (NTI)

AI for Business – Summer Training Program

National Telecommunication Institute (NTI) | ITIDA

June 2026 – July 2026 | 120 Hours | Score: 95.5%

Completed 90 hours of technical training and 30 hours of freelancing training.

Successfully completed the AI for Business summer training program with a final score of 95.5%.

SKILLS

Programming Languages: Python,SQL

AI & Machine Learning: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, LLM Integration, Multi-Agent Systems, Prompt Engineering, Multimodal AI, Zero-Shot Learning

Computer Vision: YOLO, ByteTrack, OpenCV, MediaPipe, CLIP, X-CLIP, DeepFace, Pose Estimation, Object Tracking, Emotion Recognition

Data Science: pandas, NumPy, Data Mining, Exploratory Data Analysis, Feature Engineering, Classification, Clustering, Decision Trees, KNN, K-Means, Apriori

NLP & Audio: Whisper, Speech-to-Text, Text Analysis, Prompt Optimization

Application Development: React, React Native, Expo, Streamlit, PySide6, Tailwind CSS, Browser Extension Development

Data Visualization & Reporting: Plotly, Matplotlib, Seaborn, PDF/Word/PowerPoint Report Generation

AI Engineering: LLM Grounding, Output Validation, Model Evaluation, Confidence Calibration, Privacy-Preserving AI, Bilingual Arabic–English AI Systems

Testing & Tools: Git, GitHub, GitHub Actions, pytest, Vitest, Playwright, CI/CD

PROJECTS

Buzz — Privacy-First Speech Intelligence PWA

JavaScript, Web Audio API, ONNX Runtime Web, Silero VAD, WeSpeaker, Web Workers, IndexedDB

- Built a bilingual, offline-first PWA that measures billable speaking time using neural voice activity detection running entirely on-device, reducing false detection from 41.9% to 0% in a targeted noise test.
- Implemented speaker verification and real-time diarization using voice embeddings, adaptive voiceprints, cosine-similarity clustering, automatic room calibration, and cross-session identity persistence—with zero audio sent to external servers.

Live Demo: <https://mohamed-elrady.github.io/Buzz/> 

Bodalyze — Multimodal Body-Language Analysis Platform

Python, PyTorch, YOLO11, ByteTrack, CLIP, X-CLIP, DeepFace, MediaPipe, Whisper

- Engineered an end-to-end multimodal pipeline combining pose estimation, multi-person tracking, vision-language models, facial analysis, speech transcription, and voice features.
- Converted frame-level predictions into timestamped behavioral events using confidence smoothing, per-person calibration, and hysteresis segmentation; delivered bilingual reports, privacy redaction, and temporal-IoU evaluation.

InsightLab — Human-in-the-Loop AI Analytics Platform

Python, Multi-Agent Systems, LLMs, pandas, Streamlit, Plotly, CrewAI

- Built a multi-agent analytics platform that profiles CSV/Excel data, requests business context before ambiguous decisions, and stores user knowledge as reusable, testable rules.
- Implemented deterministic analytics with numeric grounding, bootstrap significance testing, confounding detection, validated natural-language query plans, and bilingual PDF, Word, and PowerPoint reporting.

Prompt Rerank — AI Prompt Optimization Browser Extension

TypeScript, React, WXT, LLM APIs, Zod, Tailwind CSS, Vitest, Playwright

- Developed a Manifest V3 extension that locally scores prompts, identifies quality issues, and provides 17 intent-preserving optimization actions across major AI platforms.
- Integrated streaming support for OpenAI, Anthropic, Gemini, OpenRouter, DeepSeek, Ollama, and LM Studio within a privacy-first architecture backed by 108 passing tests and Playwright E2E coverage.

MUMAI — AI-Powered Child Development Tracker

React Native, Expo, TypeScript, Anthropic API, AsyncStorage

- Developed a bilingual Arabic–English application for tracking developmental milestones from birth to age five across motor, language, cognitive, social, and emotional domains.
- Built a grounded conversational assistant that maps parental questions to structured milestone data, supports optional LLM integration, stores progress locally, and runs across mobile and web.
- **Live Demo:** <https://mohamed-elrady.github.io/MUMAI/> 

CERTIFICATES

AI for Business – Summer Training Program

National Telecommunication Institute (NTI) | ITIDA

McKinsey Forward Program

McKinsey & Company

Completed the McKinsey Forward Program, developing practical skills in adaptability, problem-solving, effective communication, and digital and AI-related capabilities for the future workplace.

LANGUAGES

Arabic
(native)

English
(professional working proficiency)